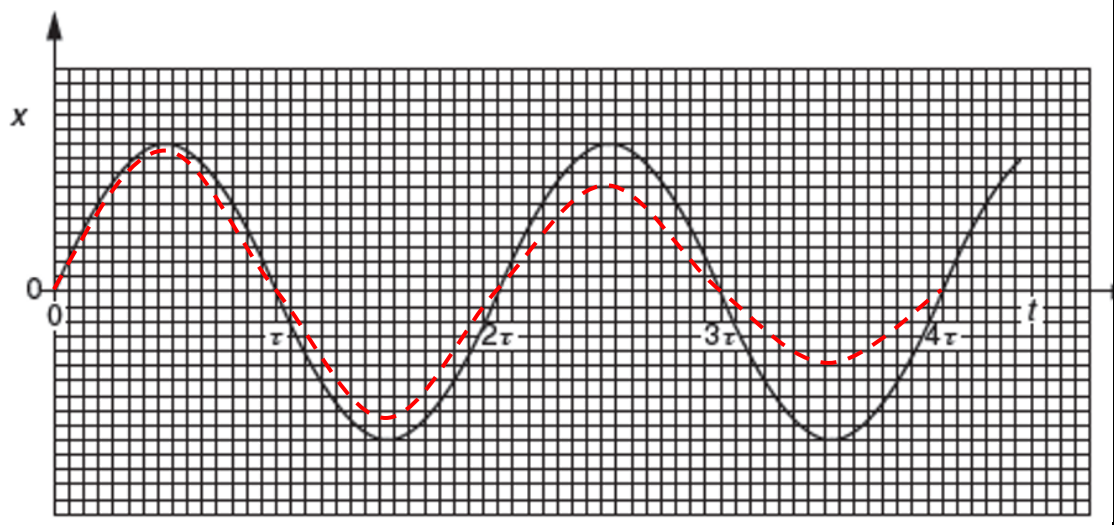
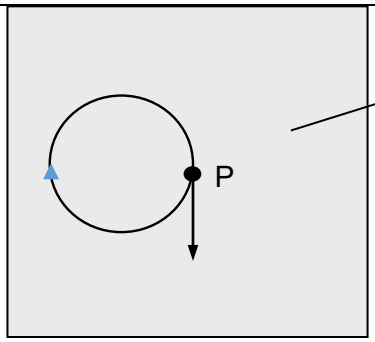
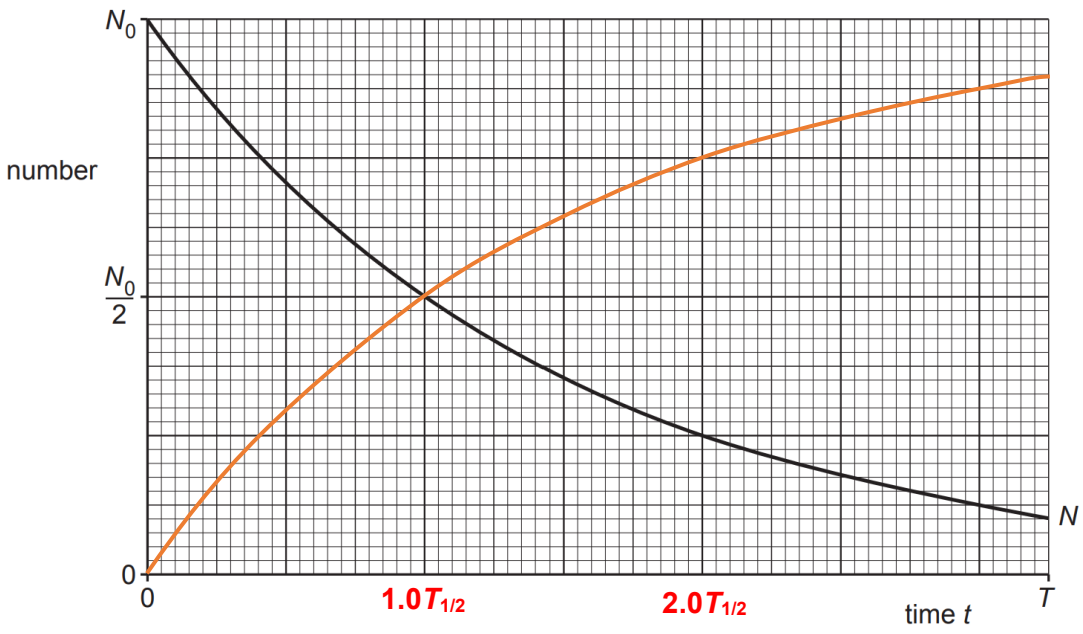
Anderson Serangoon Junior College 2026 H2 Physics Prelim Mark Scheme

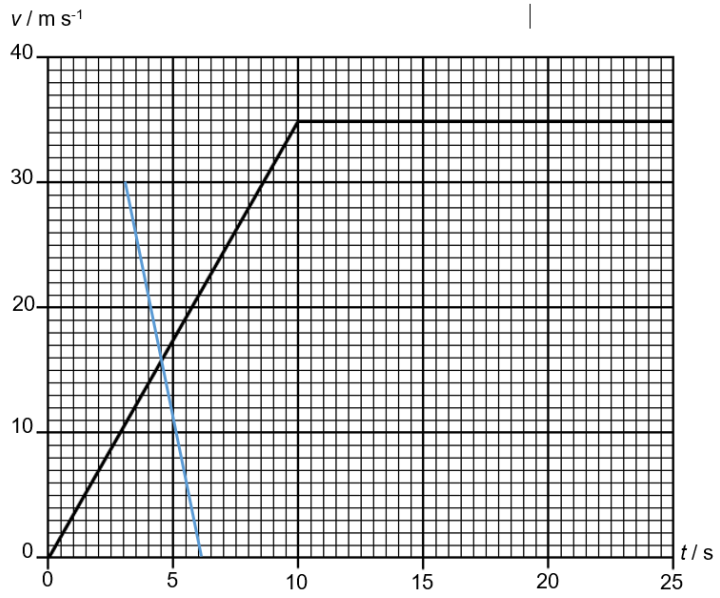
Paper 3 (75 marks)

1a	unit of pressure = (kg m s^{-2}) / m^2 = kg m^{-1} s^{-2} Unit of pressure is expressed in terms of the product and quotient of base units kg, m and s.	B1 B1
1b	$\frac{\Delta Q}{Q} = 4\frac{\Delta r}{r} + \frac{\Delta p_1 + \Delta p_2}{p_1 - p_2} + \frac{\Delta L}{L}$ $\frac{\Delta Q}{Q} = 4\frac{(0.04)}{(2.07)} + \frac{2+1}{(127-105)} + \frac{8}{200}$ = 0.25	C1 C1 A1
2a	Product of the force and the perpendicular distance from the point.	A1
2bi	Taking moments about Q (moment due to weight =) 1.2 × 270 × 9.81 (moment due to post =) 1800 × (1.6 / cos θ) 1.2 × 270 × 9.81 = 1800 × (1.6 / cos θ) θ = 25° *Alternative solution* moment due to weight = $(270 \times 9.81\cos\theta)(\frac{1.2}{\cos\theta})$ moment due to post = $1800(\frac{1.6}{\cos\theta})$ OR moment due to weight = $(270 \times 9.81\cos\theta)(1.2\cos\theta)$ moment due to post = $1800 \times 1.6\cos\theta$	C1 C1 A1
2bii	Vertical component of F = 1800 cos 25° = 1631 N Weight = 270 x 9.81 = 2649 N For vertical equilibrium (or net force zero in vertical direction), R must have a vertical component R is not horizontal OR Taking moments about CG, there is anticlockwise moment due to F, so R must provide clockwise moment R is not horizontal OR By drawing the lines of action of the three concurrent forces they must meet at a point (must indicate clearly on diagram) R is not horizontal	 M1 A1 (M1) (A1) (M1) (A1)

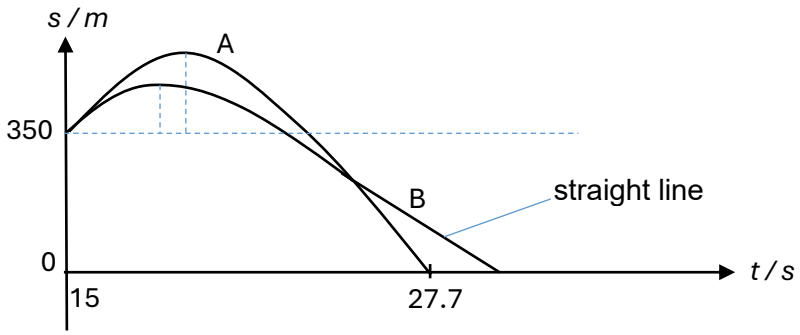
3ai	mg(Δ)h = 64 × 9.81 × 120 = 75 340.8 J = 75 kJ	A1
3aii	Elastic potential energy = area under F-h graph = 0.5 × 1800 × (120 – 37) = 75 000 J = 75 kJ	C1 A1
3b	Gravitational potential energy store has been fully transferred to / equal to elastic potential energy store Kinetic energy store is zero so <u>speed is zero</u>.	B1 B1
4ai	*g* and *r* are constants, so *a* is proportional to *x*. The <u>negative sign</u> shows *a* and *x* are in opposite directions.	B1 B1
4aii	$\omega^2 = \frac{g}{r}$ <u>and</u> $\omega = \frac{2\pi}{T}$ $\omega^2 = \frac{9.81}{0.28} = 35$ $T = \frac{2\pi}{\sqrt{35}} = 1.06$ s Time interval = half a period = *T*/2 = 1.06/2 = 0.53 s	 C1 C1 A1
4b	 period is constant, displacement decreases successive decrease in peak	 B1 B1 B1
5ai	Change in moment due to W = *mg* × (Δ)*L* = $1.3 \times 10^{-4} \times 9.81 \times 2.6 \times 10^{-2}$ = *3.32* x 10^{-5} = *3.3* x 10^{-5} N m	 A1

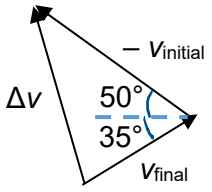
5aii	Change in moment due to W = Change in moment due to magnetic force on QL $3.3 \times 10^{-5} = B \times (\Delta)I \times L \times x$ $3.3 \times 10^{-5} = B \times 1.2 \times 0.85 \times 10^{-2} \times 5.6 \times 10^{-2}$ B = 0.058 T	C1 A1
5bi	 Circular path with diameter of circle perpendicular to velocity at P. Direction of path is clockwise.	B1 B1
5bii	magnetic force provides the centripetal force $Bev = mv^2 / r$ $r = mv / Be$ $= (9.11 \times 10^{-31})(2.9 \times 10^{7}) / (1.5 \times 10^{-3}) (1.60 \times 10^{-19})$ = 0.110 m	M1 A1
5biii	(With B and E field into the plane of the paper, the path of the electron will be) a <u>helix</u> <u>with an increasing pitch/vertical separation</u> and moving <u>out of the plane of the paper</u>.	B1 B1
6a	The <u>changing/alternating</u> current in <u>coil C</u> produces a <u>changing/alternating magnetic flux density (*B*)</u> (since magnetic flux density is directly proportional to the current). Hence, <u>coil D experiences a changing magnetic flux linkage</u> since the direction of the B field is non-parallel to the plane/area of coil D. By Faraday's law, an e.m.f. is induced in coil D.	B1 B1 A0
6b	t = 5.0 ms, 15.0 ms or 25.0 ms	A1
6c	Since the <u>gradient of the graph/rate of change of current</u> at this point is <u>the largest</u>, (it corresponds to the greatest rate of change magnetic flux density/magnetic flux linkage.) Hence, induced e.m.f. is a maximum.	A1
6di	change in magnetic flux density = $B_f - B_i$ $= 1.30 \times 10^{-3} (I_f - I_i)$ $= 1.30 \times 10^{-3} (0 - (-2.4))$ $= 3.12 \times 10^{-3}$ $= 3.1 \times 10^{-3}$ T	M1

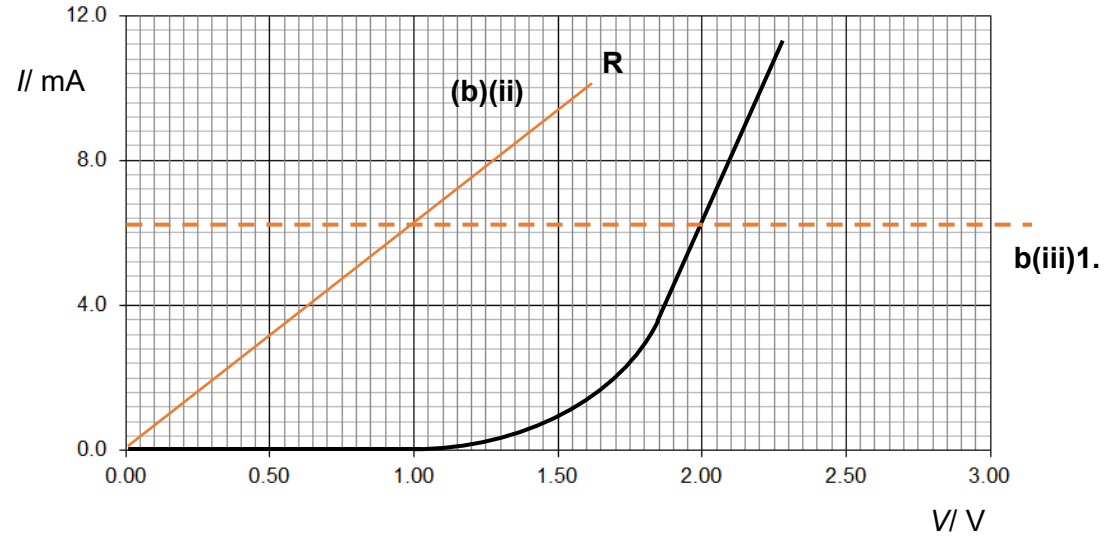
	$\text{average } \varepsilon = \left\lvert\frac{\Delta N\phi}{\Delta t}\right\rvert = \left\lvert\frac{N(\Delta B)A}{\Delta t}\right\rvert$ $= \frac{50(3.1\times10^{-3})(1.05\times10^{-3})}{5.0\times10^{-3}}$ $= 0.032 \text{ V}$	C1 A1
6dii	From t = 10 ms to 15 ms, <u>current in coil C is decreasing</u>. The (magnitude of) <u>magnetic flux linkage through coil D is decreasing</u>. To oppose the decrease in flux, coil D induces a current that <u>produces a magnetic field in the same direction as the magnetic field of coil C.</u> Hence, the current in coil D flows in the same direction as coil C.	M1 M1 A0
7ai	$\Delta N / \Delta T$	B1
7aii	$\Delta N / N$	B1
7aiii	$\Delta N / (N \Delta T)$	B1
7bi	mass difference = mass of Po – mass of Pb – mass of He = 207.981245u – 203.973043u – 4.002603u = 5.599 x 10^{-3} u energy = mass difference x c^2 = (5.599 x 10^{-3})(1.66 x 10^{-27})(3.0 x 10^{8})^2 = 8.365 x 10^{-13} J = (8.365 x 10^{-13} x 10^{-6}) / 1.6 x 10^{-19} = 5.228 MeV	 A1 C1 A1
7bii	**1.** kinetic energy (of recoil) of the lead (nucleus) **2.** there may be γ radiation	B1 B1
7ci	Mass number = 140 and atomic number = 58	A1
7cii		

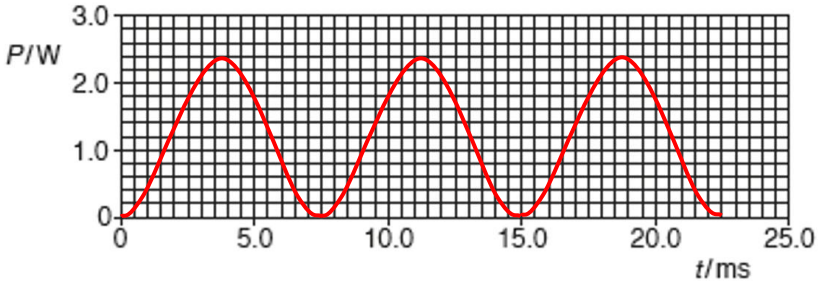
	1.0 $T½$ shown at ½ N_0 and 2.0 $T½$ shown at ¼ N_0 Line starting at (0, 0) and reaching points (T, $N_0 - N$), (1.0$T½$, ½ N_0)	B1 B1
7d	Activity of 4 g of fresh volcanic rock = 36 ÷ 3 = 12 Bq Using $A = A_0 e^{-\lambda t}$, $5 = 12e^{-(\frac{\ln 2}{7.8\times10^7})t}$ $\ln(\frac{5}{12}) = -(\frac{\ln 2}{7.8\times10^7})t$ $t = 9.9\times10^7$ years	C1 A1
8a	Gradient = (35.0 – 0.0) / (10.0) = 3.50 m s^{-2}	A1
8bi	A straight line graph with gradient about – 10, v = 30 m s^{-1} at t = 3.0 s, v = 0 at t = 6.0 s. (see 8bii for the graph)	A1
8bii	Min separation occurs when velocity of phone v_p = velocity of boy (balloon) v_b Intersection at t ~ 4.5 s  Separation = distance by boy – distance by phone = ½(3.50)(4.5)^2 – [(30)(1.5) – ½ (9.81)(1.5)^2] = 1.5 m OR separation = distance by boy – distance by phone = ½(4.5)(16.0) – ½(1.5)(30+16.0) (Areas estimated from graphs) = 1.5 m (accept 0.75 to 1.5 m) OR velocity of phone = velocity of boy	M1 C1 A1 (M1)

[Turn Over

	$u_p - g(t-3.0) = u_b + at$ $30 - (9.81)(t-3.0) = 0 + (3.50)t$ t = 4.465 = 4.47 s Separation = distance by boy – distance by phone = ½ (3.50)(4.47)^2 – [(30)(1.47) – ½ (9.81)(1.47)^2] = 1.5 m	(C)1 (A1)
8ci	Taking downwards as positive, $s_y = u_y t + \frac{1}{2} g t^2$ 0.5(35)(5+15) = − 35t + ½ (9.81)t^2 ➔ t = 12.74 = 12.7 s	C1 A1
8cii & ciii	 c)ii) smooth (parabolic) curve <u>symmetrical</u> about the highest point t = 15 s, s = 350 m & t = 27.7 s, s = 0 m **c)iii)** Initially a curve with the <u>same angle</u> as A but reach a <u>lower max</u> height at <u>an earlier time</u> than A and <u>asymmetry</u> about the highest point must be correct (longer time to return to the same height). Final section of the sketch is a straight line graph Same initial s & reaches ground later than that in (b)(ii)	 B1 B1 B1 B1 B1
8di	Since the floor is frictionless, there is <u>no horizontal force</u>. Therefore, linear <u>momentum</u> is <u>conserved</u> in the <u>horizontal</u> direction. COLM in horizontal direction m (11.5 cos 50°) = m v cos 35° v = 9.02 = 9.0 m s^{-1} **OR** Since the floor is frictionless, there is <u>no horizontal force</u>. Therefore, there is <u>no horizontal acceleration</u> and <u>horizontal velocity</u> remains <u>unchanged</u>. 11.5 cos 50° = v cos 35° v = 9.02 = 9.0 m s^{-1}	M1 A1 M1 A1

8dii	$\Delta v = \sqrt{(11.5)^2 + (9.02)^2 - 2(11.5)(9.02)\cos 85°}$ $= 13.98 = 14$ m s^{-1}. Diagram for reference only:  	M1 A1
8diii	Let F be the contact force from the ground on the ball. $F - mg = m\Delta v/\Delta t$ $F = (0.500)(13.98)/(0.20) + (0.500)(9.81)$ $= 40$ N	C1 A1
8div	There is work done against friction in horizontal direction, (so some kinetic energy is transferred to internal store). The ball will rebound with a smaller (horizontal) speed.	B1 B1
9a(i)	$R = \frac{\rho l}{A}$ $= \frac{5.0\times10^{-7}\times2.0}{3.3\times10^{-7}}$ $= 3.03\ \Omega$ $= 3.0\ \Omega$	 A1
9aii	*Current through wire* = $\frac{1.5}{3.0+0.75}$ $= 0.40$ A $I = nAqv$ v = $\frac{0.40}{8.5\times10^{28}\times3.3\times10^{-7}\times1.6\times10^{-19}}$ $= 8.91\times10^{-5}$m s^{-1}	C1 A1
9aiii	When the length of the wire is doubled, the <u>resistance</u> $R(=\frac{\rho l}{A})$ <u>of the second wire is twice</u> that of the first wire. Since $I = \frac{E}{R+r}$, <u>current through the second wire decreased</u>. Using $v = \frac{I}{nAq}$, Since the number density (as the material and temperature are the same), cross-sectional area and the charge of charge carrier (electron) are unchanged, <u>the average drift velocity of the electrons decreased.</u>	M1 M1 A1
9bi	*Using* $R = \frac{V}{I}$ = 2.25/0.0108 = 208 Ω	 M1 A1

9bii	 Line R should be a straight line passing through (0,0) and (1.60 V, 10.0 mA)	B1
9biii 1.	Draw a horizontal line on Figure 9.2 such that the pd across 160 Ω and LED1 adds up to 3.0 V. I = 6.2 mA	B1 A1
9biii 2.	With two LEDs, effective resistance of the circuit is lowered, leading to higher current from battery.With higher current, the potential difference across the 160Ω resistor is larger, leaving less potential difference across the LEDs. By $P = \frac{V^2}{R_{LED}}$,the power of the LED will be lower OR With two LEDs in parallel, effective resistance between the LEDs is lowered. By Potential Divider Principle, the potential difference across each LED is also lowered. By $P = \frac{V^2}{R_{LED}}$,the power of the LED will be lower OR From the graph, it can be seen the current (of the LED) decreases with potential difference, so the power (P=VI) dissipated in each LED is lower, so brightness is dimmer.	M1 M1 A1 (M1) (M1) (A1)
9ci	$V_s / V_p = N_s / N_p$ N_s = 52/150 x 1200 = 416 turns	M1 A1
9cii	$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{52}{\sqrt{2}}$ = 36.8 V mean power = $\frac{V_{rms}^2}{R} = \frac{(36.8)^2}{R}$ $R = \frac{(36.8)^2}{1.2}$ = 1100 Ω	C1 A1

9ciii	sine-squared graph with troughs at zero power and at least 3 ‘cycles’ peak power = 2.4 W and correct period 	B1 B1